Geometrically Inductive Self-Supervised Learning for EEG

A Design Study of Spatial Attention Across Sessions, Subjects, and Montages

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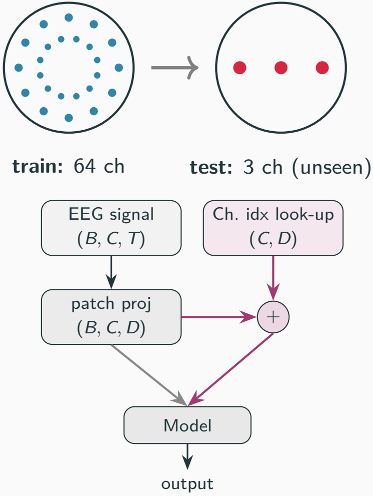
Problem: EEG is non-stationary – and the geometry is what changes

EEG varies on **three geometric axes**:

- **Session** — cap shifts day to day. *(motivates)*
- **Subject** — head sizes differ. *(motivates)*
- **Montage** — 2 to 256 electrodes in different layouts. ← **what I test**

Codex models (LaBraM, EEGPT): identity-indexed embeddings — **cannot even run** on an unseen layout.
Channel-independent (BIOT): no spatial structure at all.

Neither uses scalp geometry as an *input*.



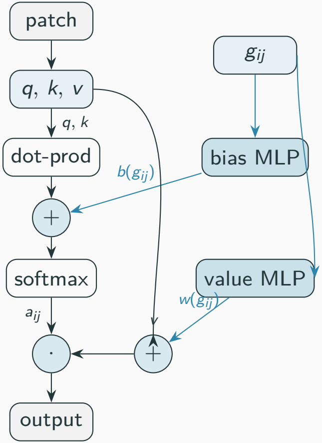
Solution: spatial attention conditioned on g_{ij}

$$g_{ij} = \left[\, p_i, \quad p_j, \quad p_i - p_j, \quad \|p_i - p_j\| \,\right] \in \mathbb{R}^{10}$$

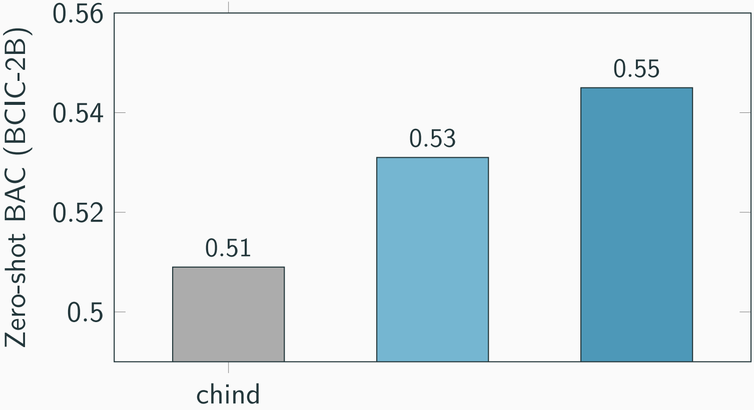
Example: $g_{\mathtt{C3}\to\mathtt{Cz}}$ (normalized coords)

p_i (C3)	(x, y, z)	$(-0.71,\ 0.00,\ 0.59)$
p_j (Cz)	(x, y, z)	$(\ \ 0.00,\ 0.00,\ 0.95)$
$p_i - p_j$	disp.	$(-0.71,\ 0.00,\ -0.36)$
$\|p_i - p_j\|$	dist.	0.80

$3+3+3+1 = 10$ dims. **pos_only** ablation keeps just $[p_i, p_j] \in \mathbb{R}^6$.



Headline: at scale, geometric beats chind on zero-shot transfer



78 PhysioNet + 78 Sleep-EDFx pretrain $\rightarrow$ zero-shot to BCIC-2B

Conclusion: a clean positive, with clear edges

What I found:

- Geometry-conditioned attention **transfers zero-shot** to an unseen montage and **beats chind significantly** once given enough data.
- Robust to descriptor reduction (pos_only also significant).

Honest limits:

- Significant on **one** held-out montage (BCIC); small-corpus tests directional but underpowered (9-subject ceiling).

Future work: second distinct-montage held-out (e.g. 128 ch); larger corpus; implement the codex baseline.

Did the results match the proposal's predictions?

Predicted vs. observed

	Predicted	**Observed**
Direction (3 montages)	geom +	**geom + (3/3)**
Powered transfer	geom ++	**geom ++, sig.**
Across all montages	hoped	1 powered; rest future

`github.com/tianxin-scu/geometric-eeg-ssl`

Key References

- Wang et al. **EEGPT**. NeurIPS 2024 – training recipe (momentum + reconstruction).
- Grill et al. **BYOL**. NeurIPS 2020 – momentum-encoder alignment.
- He et al. **MAE**. CVPR 2022 – masked-patch reconstruction.
- Hamilton, Ying, Leskovec. **GraphSAGE**. NeurIPS 2017 – inductive vs. transductive vocabulary.
- Veličković et al. **GAT**. ICLR 2018 – attention over relational features.
- Jiang et al. **LaBraM**. ICLR 2024; Yang et al. **BIOT**. NeurIPS 2023 – codex / channel-independent baselines.
- Schalk et al. 2004 – PhysioNet MI; Tangermann et al. 2012 – BCIC-2B; Kemp et al. 2000 – Sleep-EDF.

Full reference list and per-experiment details in the project report.